Example 5.2 The mean of 9 observations is 24. Find the sum of the 9 observations.

Solution

$$\text{Arithmetic Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

$$\text{Thus, } 24 = \frac{\text{Sum of all observations}}{9}$$

$$\text{Sum of all observations} = 9 \times 24 = 216 .$$

Example 5.3 The mean age of 15 teachers in a school is 42. The ages of the teachers are 35, 42, 48, X, X+8, 40, 43, 50, 46, 50, 37, 32, 38, 41, 40 (in years). Find the value of X and unknown ages of the two teachers?

Solution

$$\text{Mean} = \frac{\text{Total ages of teachers}}{\text{Number of teachers}}$$

$$42 = \frac{35+42+48+X+(X+8)+40+43+50+46+50+37+32+38+41+40}{15}$$

$$\frac{550+2X}{15} = 42$$

$$550+2X = 42 \times 15$$

$$= 630$$

$$2X = 630 - 550$$

$$2X = 80$$

$$X = \frac{80}{2}$$

$$X = 40$$

Therefore, the age of the teacher (X) is 40 and the age of the another teacher (X+8) is 40+8=48.

Example 5.4 If the mean of the following numbers is 38, find the value of x.
48, x, 37, 38, 36, 27, 35, 34, 38, 49, 33.

Solution

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

$$38 = \frac{48+x+37+38+36+27+35+34+38+49+33}{11}$$

$$38 = \frac{375+x}{11}$$

$$38 \times 11 = 375 + x$$

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